



AutoValuate
INTELLIGENCE

Explainable, damage-aware car valuation for the UAE.

One trained damage detector · one explainable price model · one agentic-RAG layer
— **every number citation-grounded.**

A group capstone · MAIB · SP Jain School of Global Management, Dubai

 AutoValuate
INTELLIGENCE

API live

 CV · explainable ML · agentic RAG

Know what your car is **really** worth.

An instant, explainable, damage-aware fair-market valuation for the UAE — backed by a trained damage detector, an explainable pricing model, and live comparable listings. Every number is traceable.

 Your vehicle
Photos optional · details required

 Drop car photos or tap to upload
Up to 8 images - scanned for damage when the CV service is live

Make	Model	Year
Toyota	Corolla	2019
Mileage (km)	Cylinders	Body type
98888	4	Sedan
Regional specs	Transmission	Fuel
GCC	Automatic	Petrol
City	Seller	
Dubai	Owner	

Value my car

 Your valuation appears here

Fill in the details and hit "Value my car". The full reasoning trace streams live, then the results build in.

THE TEAM

Four builders, one hybrid system.



Krishna Mathur

AS25DXB018

Deep learning · CV detector



Yash Petkar

AS25DXB020

Valuation ML · data



Atharva Soundankar

AS25DXB021

Agentic RAG · backend



Member 4

TBC

Frontend · product

Roles show each member's primary focus; the build was collaborative across all four pillars.

THE PROBLEM

Everyone selling a car gets lowballed.

\$20.6B

UAE used-car market by 2026 — large, real, still growing

Dealers quote low

They quote below market to protect their own resale margin. You have no independent number to push back with.

Classifieds mislead

Listing sites show asking prices, not what cars actually sell for. Anchoring on them overprices your car.

Damage is guesswork

Both sides price that door dent by gut feeling. Nobody quantifies what a specific dent, at this mileage, actually costs.

A big, high-churn market going online.

\$35 . 8B

UAE used-car market by 2030 — up from \$20.6B in 2025 (~11% CAGR)

40%

of used-car sales now online, up from 25% in 2023

~2x

used-to-new sales — resale is the bigger market

Why now

The market is worth tens of billions and growing ~11% a year, and the buying journey has moved online fast. Incumbents each cover part of it — CarSwitch runs a managed inspection service, SellAnyCar buys cars outright (and is often accused of undervaluing), dubizzle is classifieds. None gives a seller an instant, transparent, damage-aware price with the reasoning shown. That's the wedge.

WHY EXISTING TOOLS FALL SHORT

Each piece exists. Nobody stitches them together.

Instant-offer and classifieds players (dubizzle / Sell Any Car, SellAnyCar.com, CarSwitch, Seez, dubicars, YallaMotor) each solve part of it. None gives a seller a transparent, damage-aware price that shows its reasoning. To do the job properly you need all three at once:



Visual damage assessment

Sees the dents, scratches, cracks — and localizes them.



Explainable pricing

Shows exactly why it reached a number, factor by factor.



Live comparable evidence

Backs the estimate with real listings you can open.

AutoValuate is the one report that carries all three — and ties every claim back to the model that produced it.

THE SOLUTION

Photos in. A trustworthy, explained number out.



No black box: the middle step is a purpose-trained CV model, a gradient-boosted price model with SHAP, and a hybrid retrieval layer — not one LLM guessing.

LET'S SEE IT WORK

This is the running app.

Everything on the next slides is a screenshot of this exact system — not a mockup. We'll switch to it live now and value a real car end to end.

Live: autovaluate.vercel.app (deploying — runs locally today)

Every number you'll see reproduces from our evaluation suite — one command, all green.

CV • explainable ML • agentic RAG

Know what your car is **really** worth.

An instant, explainable, damage-aware fair-market valuation for the UAE — backed by a trained damage detector, an explainable pricing model, and live comparable listings. Every number is traceable.

Moderate confidence

Confidence in this estimate is moderate. No photo-based damage assessment was performed, so the figure assumes market-typical condition. We recommend a professional inspection before relying on this number for a transaction. This is an automated estimate, not a certified appraisal.

No visual damage assessment was performed

Fair-market valuation

Confidence 80% confidence range

Estimated value

AED 21,894 — mid AED 33,184 — AED 50,294

Why this price — SHAP feature impact (AED)

Visual damage assessment

Computer-visual plates

No photo-based damage scan was performed for this valuation, so the estimate assumes market-typical condition. The trained YOLOv8 detector runs on the Hugging Face Space, add photos with the service live to get a condition score.

Comparable listings

Hybrid retrieval over real Dubizzle cars

5 matches

2019 Toyota Corolla

2018 Toyota Corolla

2022 Toyota Corolla

2023 Toyota Corolla

2022 Toyota Corolla

Seller report

Synthesized via template • every figure citation-grounded

Based on the details provided, 2019 Toyota Corolla has an estimated fair-market value between AED 21,894 vs and AED 50,294 vs, with a mid-point of AED 33,184 vs. This range is a calibrated 79% confidence interval vs.

The main factors behind this estimate are noOfCylinders (-6,322 AED m), year (-6,444 AED m), bodyType (-5,318 AED m). On hold-out testing the pricing model carries a median error of about 19.57% vs, so treat the mid-point as a guide, not a guarantee.

A visual damage assessment was not available for this valuation so, so the estimate assumed a market-typical condition — a professional inspection is recommended to confirm.

Comparable five listings support this range: 2019 toyota corolla 211457km at AED 31,000 c1 ; 2018 toyota corolla 171000km at AED 29,999 c2 ; 2022 toyota corolla 69000km at AED 47,500 c3 ; 2023 toyota corolla 116850km at AED 45,499 c4 ; 2022 toyota corolla 86000km at AED 47,500 c5. If the model's confidence is limited or the car has damage beyond what the photos show, a professional inspection is the safest next step before you set a final asking price.

13 numbers and 14 citations checked by the Verifier — all trace to a computed value.

AutoValuate INTELLIGENCE

Your vehicle

Details optional details required

Drop car photos or tap to upload

Up to 8 images • scanned for damage when the CV service is live

Make Toyota Model Corolla Year 2019 Mileage (km) 98888 Cylinders 4 Body type Sedan Regional specs GCC Transmission Automatic Fuel Petrol City Dubai Seller Owner

Value my car

Live reasoning trace

Each agent, streamed in real time

Intake & validation

Damage aggregation (CV)

Price model + SHAP

Comparable listings (RAG)

Report synthesis

Citation verifier

Confidence disclosure

AutoValuate Intelligence • trained damage detector • explainable pricing • agentic RAG • not a certified appraisal

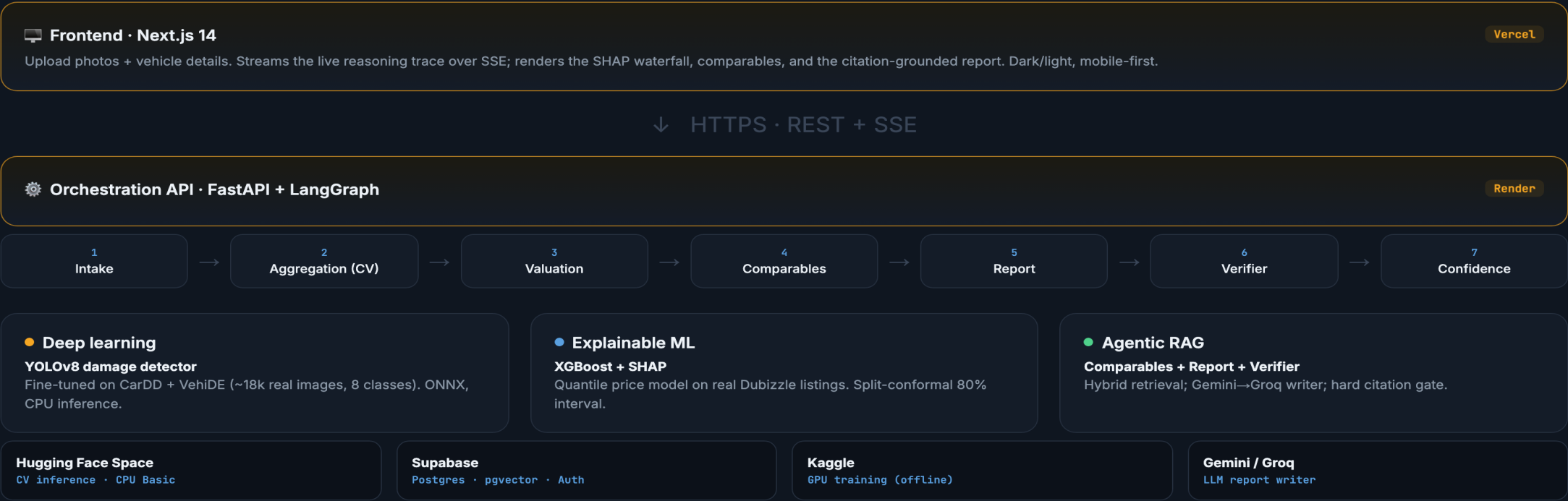
AutoValuate Intelligence • SP Jain

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A genuine hybrid — three AI systems, one product.

AutoValuate Intelligence — hybrid architecture

A trained deep-learning model + an explainable ML model + an agentic-RAG layer, on a fully free-tier stack.



A model trained to see damage — and where it is.

~18,000 real annotated images

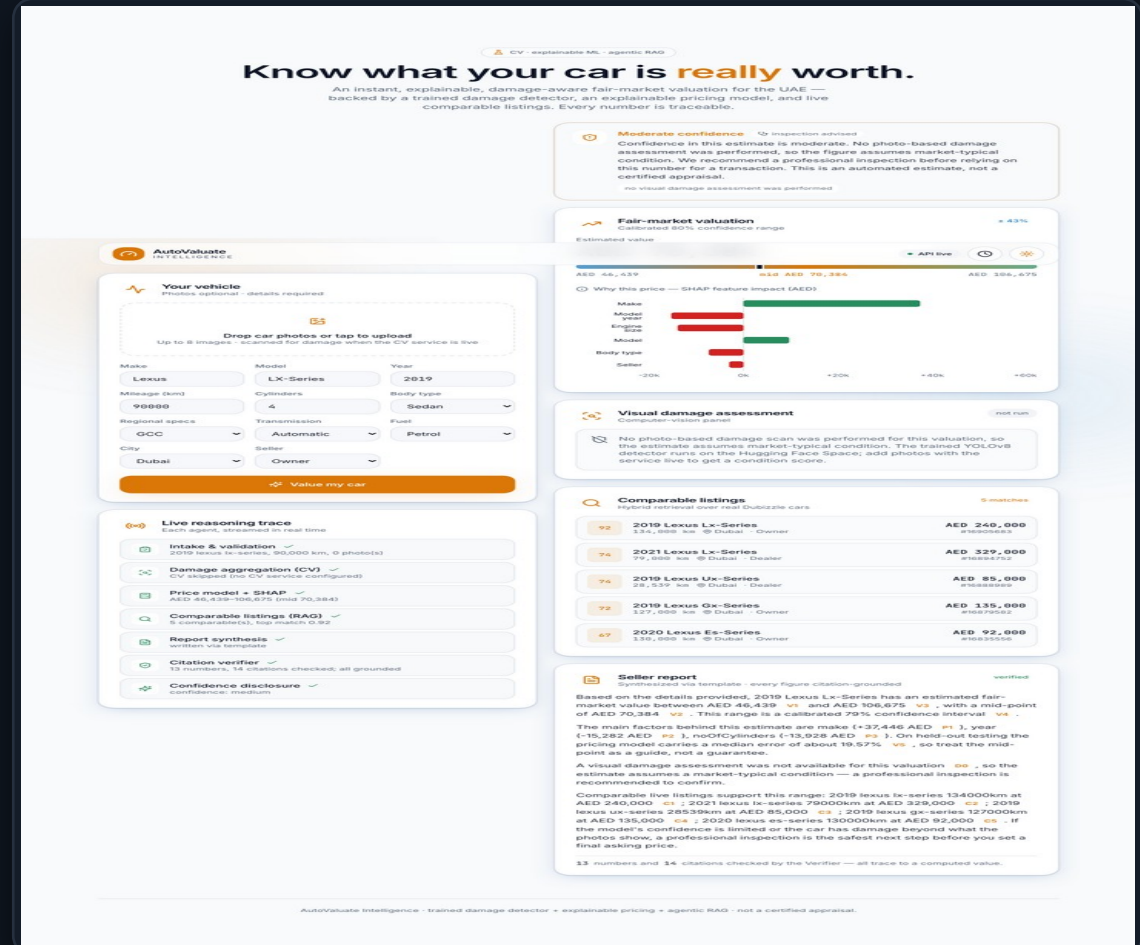
CarDD + VehiDE, unified into one 8-class schema (dent, scratch, crack, glass shatter, lamp broken, tire flat, punctured, missing part).

YOLOv8 detection, not classification

Location matters for pricing — a windshield crack costs differently than a bumper scratch. So we detect and localize, not just label.

Verified pipeline

14,437 training + 1,184 validation images unified on Kaggle; ONNX export runs CPU-only on the free Hugging Face Space.



The detector works — measured on unseen photos.

0.732

mAP@0.5 (607 held-out images)

0.58

mAP@0.5:0.95 (stricter)

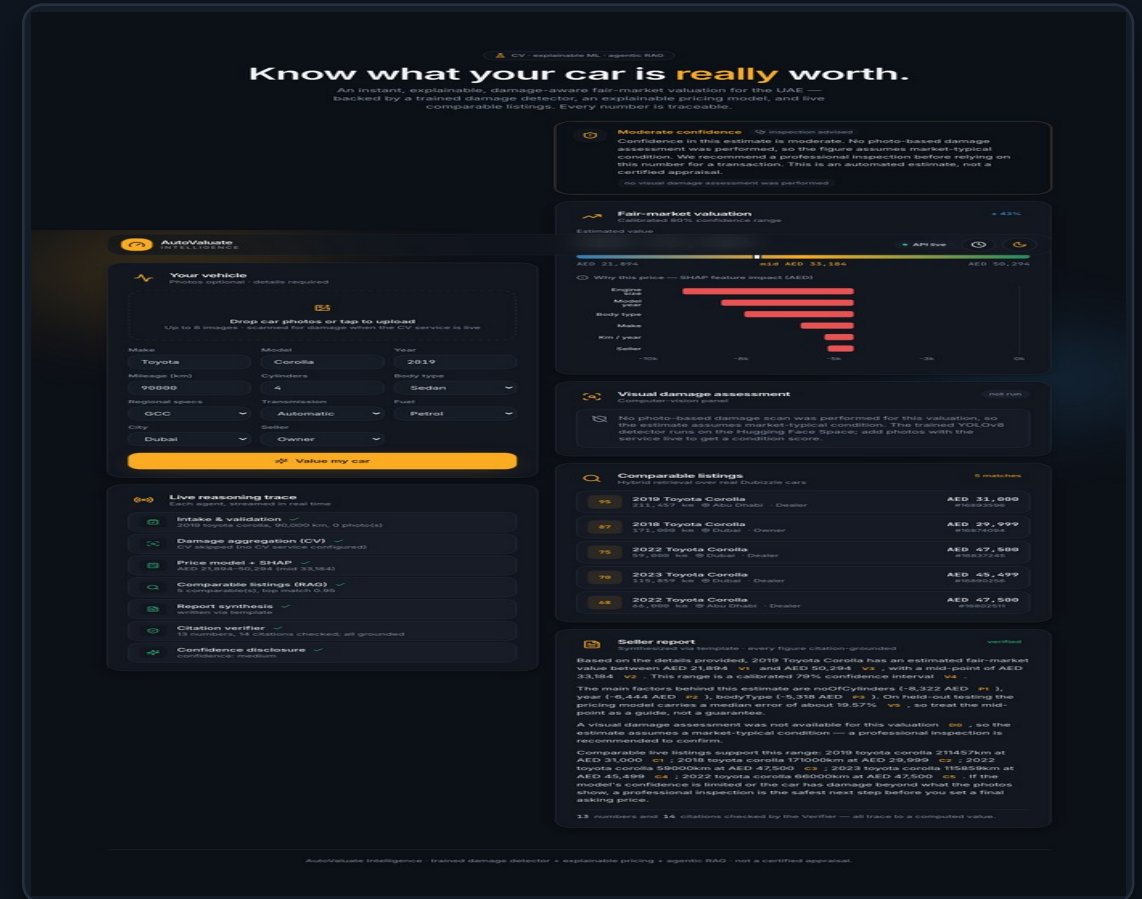
0.98

glass-shatter mAP — near perfect

0.76

mean precision · 0.69 recall

Held-out mAP (0.732) matches the training number — the detector generalises cleanly, no overfitting. Dents and scratches are solid; cracks are honestly the hardest (thin, low-contrast). Exported to ONNX and verified end-to-end: a real photo → tire-flat @0.85 → a condition score that adjusts the price.



A price you can interrogate, factor by factor.

19.6%

median error (held-out CV)

+29.3%

better than a naive baseline

0.78

honest held-out interval coverage

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SHAP directional checks pass

XGBoost quantile model on 672 real Dubizzle listings. SHAP explains each estimate; a split-conformal interval, calibrated and tested on held-out data, covers ~78% of unseen cars honestly. Mileage and age push price down, newer year lifts it — exactly as economics demands.

The screenshot displays the AutoValuate Intelligence web application. The interface is divided into several sections:

- Header:** AutoValuate INTELLIGENCE logo and status indicators (API live, clock, and refresh icons).
- Your vehicle:** A section for uploading photos (up to 8 images) and a form for vehicle details. The form includes:
 - Make: Toyota
 - Model: Corolla
 - Year: 2019
 - Mileage (km): 90000
 - Cylinders: 4
 - Body type: Sedan
 - Regional specs: GCC
 - Transmission: Automatic
 - Fuel: Petrol
 - City: Dubai
 - Seller: Owner
- Fair-market valuation:** Displays an estimated value of AED 33,184. It includes a calibrated 80% confidence range from AED 21,894 to AED 50,294, with the median value at AED 33,184. A SHAP feature impact chart shows the contribution of various factors to the price, with Engine size and Model year being the most significant.
- Live reasoning trace:** A section showing the model's reasoning process, including an intake & validation status (2019 Toyota Corolla, 90,000 km, 0 photo(s)).
- Visual damage assessment:** A section indicating that no photo-based damage scan was performed, so the estimate assumes market-typical condition.

Intake → Aggregation (CV) → Valuation → Comparables → Report → Verifier → Confidence. Each step streams to the UI over SSE, so you watch the reasoning happen.

Every AED figure and citation in the report must trace to a computed value. In testing it caught an injected AED 999,999, a fake citation, and an ungrounded percentage.

Gemini writes the report, Groq is the fallback, and a deterministic writer keeps it working with no keys at all.

AutoValuate

INTELLIGENCE

API live

Your vehicle

Photos optional - details required

Drop car photos or tap to upload

Up to 8 images - scanned for damage when the CV service is live

Make

Toyota

Model

Corolla

Year

2019

Mileage (km)

90000

Cylinders

4

Body type

Sedan

Regional specs

GCC

Transmission

Automatic

Fuel

Petrol

City

Dubai

Seller

Owner

Value my car

LIVE REASONING TRACE

Each agent, streamed in real time

Intake & validation

2019 toyota corolla, 90,000 km, 0 photo(s)

Visual damage assessment

Computer-vision panel

No photo-based damage scan was performed for this valuation, so the estimate assumes market-typical condition. The trained YOLOv8

assessment was performed, so the figure assumes market-typical condition. We recommend a professional inspection before relying on this number for a transaction. This is an automated estimate, not a certified appraisal.

no visual damage assessment was performed

Fair-market valuation

Calibrated 80% confidence range

Estimated value

AED 33,184

median error 19.57%
on 672 real listings

AED 21,894mid AED 33,184AED 50,294

Why this price — SHAP feature impact (AED)

Engine size

Model year

Body type

Make

Km / year

Seller

-10k-8k-5k-3k0k

It tells you how sure it is — and when not to trust it.

States its confidence

Every report gives the price-interval width and, when photos are provided, the detector's per-damage confidence.

Recommends inspection when unsure

Low or moderate confidence triggers a plain-English 'get a professional inspection' — it never presents a false-certain number.

Enforced, not aspirational

The disclosure contract is a test: 90 checks across 18 vehicles, zero failures. It never claims to be a certified appraisal.

The screenshot displays the AutoValuate Intelligence interface. On the left, there's a form to input car details: Make (Lexus), Model (LX-Series), Year (2019), Mileage (90000 km), Cylinders (4), Body type (Sedan), Regional specs (GCC), Transmission (Automatic), Fuel (Petrol), City (Dubai), and Seller (Owner). A prominent orange button labeled 'Value my car' is at the bottom of the form.

On the right, a list of comparable vehicles is shown with their estimated values and confidence scores:

- 2021 Lexus Lx-Series: AED 329,000 (Confidence 74)
- 2019 Lexus Ux-Series: AED 85,000 (Confidence 74)
- 2019 Lexus Gx-Series: AED 135,000 (Confidence 72)
- 2020 Lexus Es-Series: AED 92,000 (Confidence 67)

Below the form, a 'Live reasoning trace' section provides a detailed breakdown of the valuation process:

- Intake & validation**: 2019 lexus lx-series, 90,000 km, 0 photo(s)
- Damage aggregation (CV)**: CV skipped (no CV service configured)
- Price model + SHAP**: AED 46,439-106,675 (mid 70,384)
- Comparable listings (RAG)**: 5 comparable(s), top match 0.92
- Report synthesis**: written via template
- Citation verifier**: 13 numbers, 14 citations checked; all grounded

On the far right, a 'Seller report' section provides a synthesized summary of the valuation, including a market value range (AED 46,439 to AED 106,675) and a calibrated 79% confidence interval. It also includes a visual damage assessment note and a recommendation for a professional inspection.

Free for sellers. Dealers and marketplaces pay.

Individual sellers

FREE

3 valuations / month. Builds the top of the funnel and the data flywheel.

Used-car dealers

SAAS SEATS

Bulk trade-in intake + a defensible, damage-aware price to justify offers. The paying core.

Marketplaces / OEMs

API LICENCE

A 'verified estimate' badge and damage API on their own listings. Highest-value, B2B.

MVP proves the engine; the dealer tier is the first revenue and the near-term focus.

Zero cost — with the engineering judgment to prove it.

Vercel

Frontend (Next.js)

Render

Orchestration API

Hugging Face

CV inference · CPU Basic

Supabase

Postgres · pgvector · Auth

Kaggle

GPU training (offline)

Gemini / Groq

LLM report writer

Render, not Railway (Railway has no real free tier in 2026). **The CV model runs on Hugging Face** — 512MB can't hold it. **A scheduled ping** keeps Supabase awake, so the link never dies.

EVALUATION SUMMARY

Every number here reproduces from one command.

19.6%

valuation median error

1.00

comparables same-make P@5

1.000

report faithfulness

0.000

faithfulness — hallucinated control

90 / 90

confidence-contract checks

0.732

CV detector mAP@0.5 (held-out)

CV detector: mAP@0.5 = 0.732 on 607 held-out images (glass-shatter 0.98) — the deep-learning pillar, measured the same honest way.

What it does not do — named on purpose.

CPU-speed CV inference

The detector runs on CPU (a few seconds/image) and, to fit the 512MB free tier, is opt-in in the deployed API — full-speed on a paid dyno.

No accounts yet

Auth, saved history per user, and multi-tenant isolation are the next build — today it's a single-session tool.

Modest tabular set

672 real listings is small, so the price interval is genuinely wide — which is why the range is disclosed, not hidden.

No accident-history feed

There's no free UAE accident-record API, so the estimate can't see undisclosed history — hence 'get an inspection.'

From capstone to first paying dealer.

Next 30 days

Finish CV training, add Supabase Auth + saved history, ship the ONNX query-embedder so it runs in Render's free RAM, deploy all three URLs.

60–90 days

Dealer workspace: bulk intake, PDF reports, per-seat quotas. Pilot with one Sharjah/Dubai used-car dealer for real feedback.

6 months

Verified-estimate API for a marketplace, price-trend alerts, and a Dubizzle auto-fill extension — the retention and B2B layers.

One project, three things proven.

- **Deep learning** a trained, evaluated CV damage detector
- **Explainable ML** XGBoost + SHAP, a price you can interrogate
- **Agentic RAG** orchestration with a hard citation gate

T e a m

Krishna Mathur · Yash Petkar · Atharva Soundankar · +1

Thank you — happy to open any part of it live.